

CANAL CONDITION ASSESSMENT

Sage Creek Main Canal — Reach 7
Station 142+00 to 166+00

Magic Valley, Twin Falls County, Idaho

Prepared for:

Sage Creek Irrigation District

P.O. Box 284, Buhl, Idaho 83316

Prepared by:

High Desert Engineering, LLC

1450 Blue Lakes Boulevard N, Suite 200

Twin Falls, Idaho 83301

HDE Project No. 2023-IR-0087

September 18, 2023

Prepared by:

T. R. Vasquez, P.E.

Water Resources Engineer

Idaho PE-14205

Reviewed by:

M. L. Brennan, P.E.

Principal Engineer

Idaho PE-9876

1.0 INTRODUCTION

High Desert Engineering, LLC (HDE) was retained by the Sage Creek Irrigation District (SCID) to perform a condition assessment of a 2,400-foot reach of the Sage Creek Main Canal designated as Reach 7 (Station 142+00 to 166+00). This reach has experienced progressive deterioration of the concrete canal lining, a right-bank slope failure, and significant seepage losses that are threatening water delivery obligations to downstream users. The assessment was performed during the non-irrigation season (September 2023) with the canal dewatered.

1.1 Canal History

The Sage Creek Main Canal was originally constructed as an unlined earthen canal by the U.S. Bureau of Reclamation in 1924 as part of the Sage Creek Division of the Minidoka Project. The canal is 12.4 miles long with a design capacity of 280 cubic feet per second (cfs) and serves approximately 42,000 irrigated acres in the Magic Valley. Reach 7 was lined with unreinforced concrete panels (4-inch nominal thickness) in 1962 under the USBR Canal Lining Program. The lining is now 61 years old and has exceeded its 40-year design life.

The canal cross-section in Reach 7 is trapezoidal with a bottom width of 14 feet, 2H:1V side slopes, and a design flow depth of 5.5 feet. The canal is in a cut section through a low ridge, with the right bank up to 12 feet above the canal invert and the left bank approximately 6 feet above invert. The canal alignment runs approximately east-west through gently rolling terrain with irrigated agriculture on both sides.

1.2 Regional Geology

The canal traverses the northern margin of the Snake River Plain in an area underlain by Quaternary basalt flows (Shoshone and McKinney Butte lava fields) at depth, overlain by wind-deposited loess (silt) and lacustrine clay deposits associated with ancestral Lake Idaho. The geologic sequence encountered in previous excavations and water well logs in the area is:

- **Loess (0 to 5-15 ft):** Wind-deposited silt (ML to CL), tan to brown, moderately stiff when dry, collapsible when wetted. Widespread in Magic Valley. Known for collapse settlement when saturated under load (Holocene to late Pleistocene).
- **Lacustrine clay (5-15 to 15-25 ft):** High-plasticity clay (CH), green-gray to olive, stiff to very stiff, highly expansive. Deposited in ancestral Lake Idaho (Pliocene-Pleistocene). Known to cause pavement heaving, foundation damage, and lining failure throughout the Magic Valley region.
- **Alluvial sand and gravel (variable, 0-5 ft thick):** Discontinuous lenses of poorly graded sand and gravelly sand (SP, GP-GM) within and between the loess and clay units. These lenses serve as preferential seepage pathways.
- **Basalt bedrock (15-30 ft):** Vesicular to massive olivine basalt of the Snake River Group. Highly variable permeability — massive zones are nearly impermeable, while vesicular zones and flow contacts can be extremely permeable (Lugeon values > 50).

2.0 FIELD OBSERVATIONS

2.1 Concrete Lining Condition

The concrete lining in Reach 7 was inspected panel by panel during the dewatered condition. The lining consists of unreinforced concrete panels approximately 12 feet wide by 12 feet long (transverse by longitudinal) with sealed construction joints. Panel thickness, measured at exposed edges and broken panels, ranges from 3.5 to 4.5 inches.

Table 2-1: Lining Panel Condition Summary (200 panels total in Reach 7)

Condition	No. of Panels	Percent of Total	Description
Good	28	14%	Intact, minor surface scaling, functional joints
Fair	52	26%	Moderate cracking, joint sealant failed, minor spalling
Poor	78	39%	Extensive cracking, panel corners broken, subgrade exposed
Failed	42	21%	Panel heaved, displaced, or collapsed; open subgrade

The most severe damage is concentrated between Sta. 148+00 and 158+00, where 60 percent of the failed panels are located. Failed panels exhibit upward heaving of 2 to 6 inches at panel centers, consistent with subgrade expansion from the underlying lacustrine clay. Several panels have been completely displaced by vertical frost heave combined with clay swell, creating gaps of 1 to 3 feet where the subgrade is directly exposed to canal flow. Freeze-thaw damage (scaling, popouts, delamination) is present on virtually all panels and has reduced the effective thickness to 2.5 to 3.0 inches in the worst areas.

The lining concrete shows classic signs of alkali-silica reaction (ASR) in approximately 30 percent of panels, including map cracking and white gel deposits. The volcanic aggregate (crushed basalt) used in the original mix is known to be alkali-reactive in the Magic Valley region. The combination of ASR weakening and freeze-thaw cycling has created a synergistic degradation mechanism that has accelerated lining failure over the past two decades.

2.2 Right Bank Slide

A rotational slope failure was identified on the right canal bank between Sta. 152+50 and 154+30 (approximately 180 feet long). The slide exhibits the following characteristics:

- **Head scarp:** 8 to 10 feet behind the bank crest, 2 to 3 feet of vertical displacement, tension cracks extending 15 feet beyond the scarp.
- **Slide mass:** The right bank has bulged approximately 18 inches into the canal prism, displacing the lining panels at the bank toe. The slide mass is approximately 180 feet long, 20 feet wide (bank crest to canal invert), and an estimated 8 to 12 feet deep.
- **Failure surface:** The failure appears to daylight at the canal invert, where lining panels have been pushed upward and cracked by the slide mass. A greenish-gray clay seam (likely lacustrine CH clay) is visible at the base of the exposed head scarp.
- **Seepage:** Wet areas and standing water were observed in the tension cracks and at the base of the head scarp, indicating that irrigation-season canal seepage has saturated the bank materials. A small seep (estimated 0.5 gpm) was observed at the slide toe in the canal invert.

The slide is attributed to: (1) saturation of the loess bank materials by canal seepage through the failed lining, causing loss of the apparent cohesion that provides dry-season stability; (2) the presence of a low-strength clay

layer at or near the slide base; and (3) potential collapse settlement of the loess upon wetting, which creates downward movement and tension cracks at the crest.

2.3 Seepage Losses

Seepage losses in Reach 7 were measured by SCID operations staff during the 2023 irrigation season using the inflow-outflow method. Flow measurements were taken at the upstream and downstream ends of the reach using calibrated Parshall flumes.

Table 2-2: Reach 7 Seepage Loss Measurements, 2023 Irrigation Season

Date	Upstream Flow (cfs)	Downstream Flow (cfs)	Loss (cfs)	Loss (%)	Notes
May 15	230	212	18	7.8%	Early season, low soil moisture
Jun 20	255	234	21	8.2%	Full delivery, warm
Jul 18	260	238	22	8.5%	Peak season
Aug 12	245	228	17	6.9%	Slightly reduced demand
Sep 5	180	166	14	7.8%	Late season

Average seepage loss across Reach 7 is 18.4 cfs over the 2,400-foot reach, equivalent to a loss rate of 7.7 cfs per 1,000 feet. For comparison, a well-maintained concrete lining typically limits losses to 0.5 to 1.0 cfs per 1,000 feet, indicating that Reach 7 losses are approximately 8 to 15 times higher than expected for a lined canal. At peak season flows, the 22 cfs loss represents approximately 8.5 percent of the total canal capacity, directly reducing deliverable supply to the 12,000 acres served downstream of Reach 7.

The estimated annual volume of seepage loss is 5,400 acre-feet (based on 180-day irrigation season at average loss rate), valued at approximately \$270,000 at the current SCID water rate of \$50 per acre-foot. This represents a significant economic loss to the District and its water users.

2.4 Ponding Test

A ponding test was performed in the most severely damaged sub-reach (Sta. 150+00 to 156+00, 600 feet) during the dewatered period. The sub-reach was isolated using temporary bulkheads and filled to 4.0 feet depth. Water level decline was monitored over 48 hours.

Table 2-3: Ponding Test Results (Sta. 150+00 to 156+00)

Time (hr)	Water Level (ft)	Decline (ft)	Cumulative Loss (gal)	Rate (gpm)
0	4.00	—	—	—
6	3.72	0.28	12,600	35.0
12	3.48	0.52	23,400	30.0
24	3.05	0.95	42,750	26.8
48	2.25	1.75	78,750	25.0

The ponding test indicates a seepage rate of approximately 25 to 35 gpm per 600 linear feet (declining from initial saturation rate). The higher initial rate likely reflects wetting of unsaturated loess soils, which then partially seal as the soil structure collapses. The steady-state rate (approximately 25 gpm or 0.056 cfs per 600 ft) extrapolated to the full 2,400-foot reach is consistent with the 18 cfs average loss measured during the irrigation season, confirming that the majority of seepage occurs through exposed subgrade at failed panel locations and through cracked/deteriorated panels.

3.0 CONDITION RATINGS

Table 3-1: Condition Ratings — Reach 7

Component	Rating	Trend	Risk	Recommended Action
Concrete lining (overall)	POOR	Worsening	HIGH	Full replacement in Reach 7
Lining subgrade	POOR	Worsening	HIGH	Investigate expansive clay, design treatment
Right bank stability	POOR	Active failure	HIGH	Investigate, stabilize before relining
Left bank stability	FAIR	Stable	MODERATE	Investigate subgrade for lining design
Seepage control	FAILED	Worsening	HIGH	Address through lining replacement
Canal cross-section	FAIR	Reduced by slide	MODERATE	Restore after bank stabilization

Overall Reach 7 Rating: POOR — The reach is functional but is experiencing progressive failure that will worsen with each freeze-thaw and irrigation cycle. Without intervention, the slide will continue to encroach into the canal prism, and seepage losses will increase as more panels fail. Complete rehabilitation of the 2,400-foot reach is recommended within 2 to 3 years.

4.0 RECOMMENDATIONS

4.1 Geotechnical Investigation (2024-2025)

Prior to rehabilitation design, a geotechnical investigation is required to characterize the canal subgrade, bank materials, and the right-bank slide. The investigation should include the following minimum scope:

- Four exploratory borings: two at the canal invert (slide area and heave area), one at the right bank slide head scarp, and one at the left bank in a stable area for baseline comparison.
- Laboratory testing: Atterberg limits, grain-size analysis, moisture-density relationships, swell pressure and free swell tests (ASTM D4546), collapse potential tests (ASTM D5333), direct shear tests, permeability (falling-head), and concrete core testing on removed lining panels.
- Slug tests: In-situ permeability testing in two borings to characterize seepage pathways.
- Engineering analysis: slope stability of the right bank (pre- and post-remediation), subgrade treatment alternatives for expansive clay, lining design comparison (geomembrane, shotcrete, reinforced PCC, geomembrane-under-concrete composite).

4.2 Estimated Rehabilitation Costs

Based on comparable canal lining projects in the Magic Valley region, preliminary cost estimates for Reach 7 rehabilitation are:

Table 4-1: Preliminary Rehabilitation Cost Estimate

Item	Quantity	Est. Unit Cost	Est. Total	Notes
Geotechnical investigation	LS	—	\$65,000-\$85,000	4 borings, lab, analysis

Item	Quantity	Est. Unit Cost	Est. Total	Notes
Demolition of existing lining	28,800 SF	\$3.50/SF	\$101,000	Remove and dispose
Right bank slide repair	LS	—	\$120,000-\$180,000	Depending on method
Subgrade preparation	28,800 SF	\$4-8/SF	\$115,000-\$230,000	Depends on clay treatment
New canal lining	28,800 SF	\$12-22/SF	\$346,000-\$634,000	Depends on lining type
Mobilization, contingency	LS	15%	\$102,000-\$187,000	

Total estimated rehabilitation cost: \$849,000 to \$1,417,000, depending on lining type and subgrade treatment. The geotechnical investigation is essential to narrow this range and select the most cost-effective rehabilitation approach. SCID should consider applying for USBR WaterSMART Water and Energy Efficiency grant funding, which has supported similar canal rehabilitation projects in the region.

This assessment has been prepared for the exclusive use of the Sage Creek Irrigation District for planning purposes. Cost estimates are preliminary and subject to revision based on the results of the recommended geotechnical investigation.

T. R. Vasquez, P.E.

Water Resources Engineer

High Desert Engineering, LLC